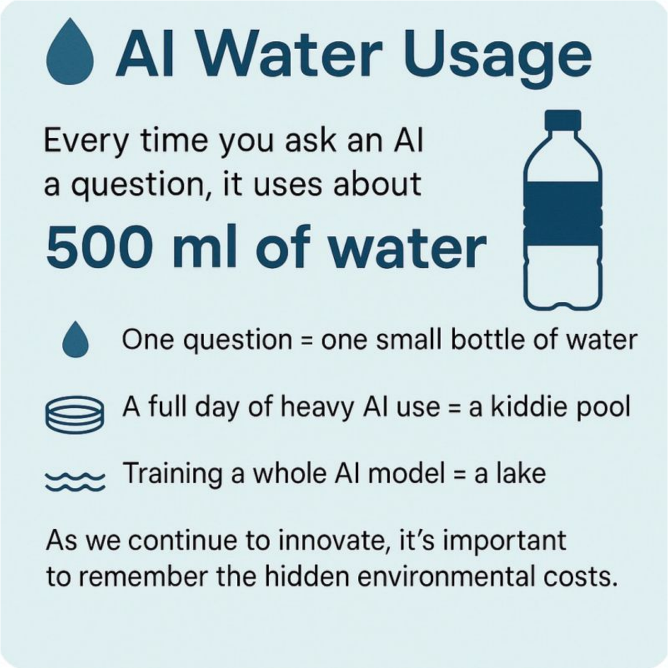
A common recent post:



Picture Credits: Google Images

- Who tracks that?
- Who audits the fleet of data centres?
- Who writes a query that tells a CEO that their infrastructure used more water than the city next door?

- **Who tracks that?**
- **Who audits the fleet of data centres?**
- **Who writes a query that tells a CEO that their infrastructure used more water than the city next door?**

It starts with databases!

- **Who tracks that?**
- **Who audits the fleet of data centres?**
- **Who writes a query that tells a CEO that their infrastructure used more water than the city next door?**

It starts with databases!

Databases **store the data** that AI models and neural networks use for training. They can efficiently handle large amounts of data, which often needs to be cleaned before the model can be properly trained.

Learning Outcomes

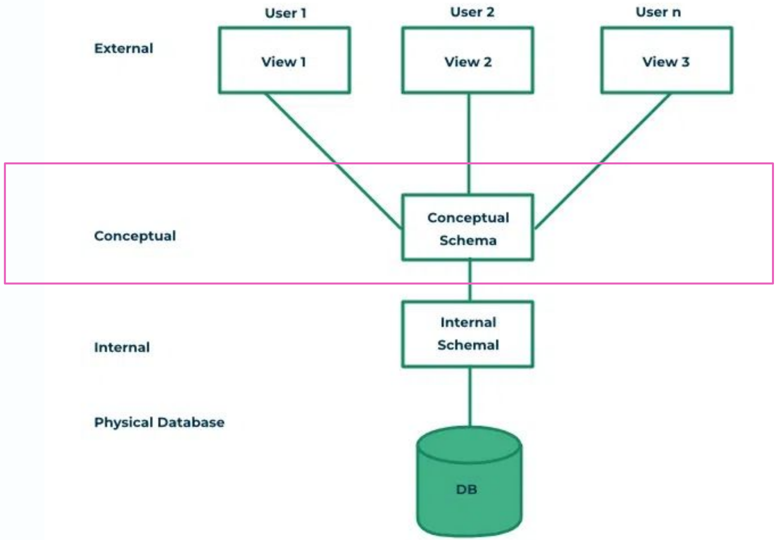
- The relational data model
- Relational Algebra
- Querying and updating databases (SQL)
- Application programming with SQL
- Integrity constraints
- Normal Forms
- Database Design
- Query Processing, Transaction Management

That sounds like a lot...

We’ll take it slowly!

We'll start here - with the conceptual schema!

By the end of this week, you'll be
introduced to designing
better databases than your friends who
still use CSVs!



Course Admin

5 Checkpoints (1 project)

Assessment and Deadlines

Description	Due Date	Weight
Project: Research Network Database		
Checkpoint 1 — ERD, single Lab	2026-05-17 (by 5:00 PM)	4%
Checkpoint 2 — Full ERD + schema + DDL	2026-05-31 (by 5:00 PM)	8%
Checkpoint 3 — SQL + RA query set	2026-06-14 (by 5:00 PM)	8%
Checkpoint 4 — Normalization	2026-07-26 (by 5:00 PM)	8%
Checkpoint 5 — Transactions	2026-08-09 (by 5:00 PM)	8%
PCRS Homework	Weekly (before each lecture)	4%
In-Person Midterm	2026-07-22 (lecture slot)	15%–20%
Final Exam	Final Exam Period	40%–45%
Total		**100%**

Resubmissions!

Project Checkpoints

All five Checkpoints form one term-long project on the **Research Network** domain. Each builds directly on the previous, improving Checkpoint *n* also improves your starting point for Checkpoint $n + 1$.

Format requirement: All written solutions must be typeset in LaTeX. Missing files, non-compiling submissions, or syntax-error-only work receives a grade of 0.

Checkpoint	Weight	Due	Deliverables	Resubmission
Checkpoint 1 ERD: single Lab	4%	May 17 5 PM	ERD for one Lab entity cluster (Researcher, Lab, Equipment). Course notation only. All cardinalities and participation constraints. Written justification ≤150 words.	1 free resubmit within **48 hours** of grade release. Must include a changelog note.
Checkpoint 2 Full network ERD + schema + DDL	8%	May 31 5 PM	ERD for all 10 entities. Relational schema with all FKs. PostgreSQL DDL must run without error on the course server.	1 resubmit within **1 week**. Only DDL/schema may be revised, ERD grade locked after first submission.
Checkpoint 3 SQL + RA query set	8%	June 14 5 PM	8 queries in SQL and RA: publications from a network · researchers on multiple projects · underfunded projects · labs with no grants · top-cited publications · expired grants · projects spanning >2 institutions · datasets in >3 publications. Include a README.	1 resubmit within **1 week**. Test cases shared post-submission so you can diff and fix.
Checkpoint 4 Normalisation	8%	July 26 5 PM	Given a denormalised "GiantResearchTable": identify all FDs, compute the minimal cover, decompose into BCNF (or justify 3NF), document all anomalies, produce final normalised DDL.	1 resubmit within **1 week**. May revise FD set or decomposition, must state what changed.
Checkpoint 5 Transactions	8%	Aug 9 5 PM	**Scenario A:** Two labs claim the same Equipment, you will write a SQL transaction pair, induce and resolve the conflict. **Scenario B:** Researcher submits to two projects with conflicting grant rules, model the schedule, draw a precedence graph, identify serializability. For each: identify isolation level and explain with 2PL or MVCC.	No resubmits for Sprint 5.

> **Grace tokens:** Each student receives 10 tokens, each worth a 2-hour extension on any checkpoint. Stack all 10 on one deadline for a 20-hour window. Once exhausted, late submissions receive a 0. Tokens are deducted automatically by MarkUs.

First Big Deadline:

- Checkpoint 1
  - Tutorial 1 (tomorrow) prepares you for this

PCRS

- [Prepare] Due next Tuesday, before lecture.
- [Practice] Due next Sunday, after lecture.

Course Philosophy: Students First

I believe the best learning happens when students feel supported, not stressed. A few principles that shape how this course is run:

Better later is still better	The 5% between the midterm and final always shifts to whichever score is higher. If you struggled on the midterm but put in the work and did better on the final, that growth is reflected in your grade. The same idea runs through resubmissions: a single bad attempt shouldn't define your outcome.
Good work takes iteration	Every Checkpoint has a resubmission window. Use the feedback, fix your work, and resubmit. Students who revise learn more than students who move on.
Each piece builds the next	Every assignment uses the same Research Network. My goal is to make sure you are not too lost in a new scenario, the context builds over time and my goal is to make sure everything connects.
Low-stakes entry	Checkpoint 1 is intentionally small and forgiving (baby assignment). The goal is to orient you.
Partial credit beats zero	If you're at risk of missing a deadline, submit whatever you have. Partial credit is always better than a zero, and the resubmission window exists for exactly this situation.
Please reach out	If something is going wrong, personally, academically, or otherwise, please talk to me. I can't always fix things but I promise to always help you find a path forward.

Midterm and Exam

Assessment and Deadlines

Description	Due Date	Weight
Project: Research Network Database		
Checkpoint 1 — ERD, single Lab	2026-05-17 (by 5:00 PM)	4%
Checkpoint 2 — Full ERD + schema + DDL	2026-05-31 (by 5:00 PM)	8%
Checkpoint 3 — SQL + RA query set	2026-06-14 (by 5:00 PM)	8%
Checkpoint 4 — Normalization	2026-07-26 (by 5:00 PM)	8%
Checkpoint 5 — Transactions	2026-08-09 (by 5:00 PM)	8%
PCRS Homework	Weekly (before each lecture)	4%
In-Person Midterm	2026-07-22 (lecture slot)	15%–20%
Final Exam	Final Exam Period	40%–45%
Total		**100%**

Tutorials

- Help prepare for checkpoints
- Attendance is not mandatory, but each lab prepares you for the checkpoint
- Qazi, the lab TA, is an experienced CSC343 TA!
    - Get help on time!

Generative AI Policy

- The course is open to it!
  - But be transparent about it.
  - Don’t be a ChatGPT wrapper… **think critically**!

Questions about admin related issues?

- Missed assignments?
- Midterm/Exam?
- Travel?

On my end:

- I will answer your questions in 48 hours!
  - I am probably not ignoring you… and just taking care of my ~~pokemon~~ thesis/travelling
  - If there are fires (MarkUs, PCRS, PSQL)
    - Don’t panic! We’ll sort it out together.

Problem 1: This Data is Broken

- What is wrong in storing it this way?
- How would you fix it?
  - Sketch or write something – anything.

model	researcher	email	lab	dataset	sz_tb	license
GPT-X	Alice Chen	alice@lab.ai	VisionLab	ImageNet-22k	1.2	CC-BY
GPT-X	Alice Chen	alice@lab.ai	VisionLab	LAION-5B	5.0	CC-BY
LLaMA-4	Alice Chen	alice@lab.ai	NLPLab	C4	0.8	ODC
LLaMA-4	Bob Nkosi	bob@lab.ai	NLPLab	C4	0.8	ODC
DiffNet	Carol Diaz	carol@lab.ai	VisionLab	COCO-2024	0.3	CC-BY-NC

Problem 1: This Data is Broken

- What is wrong in storing it this way?

Scenario 1: Alice leaves VisionLab and goes to NLP lab.
You now need to update 3 (or more) rows

model	researcher	email	lab	dataset	sz_tb	license
GPT-X	Alice Chen	alice@lab.ai	VisionLab	ImageNet-22k	1.2	CC-BY
GPT-X	Alice Chen	alice@lab.ai	VisionLab	LAION-5B	5.0	CC-BY
LLaMA-4	Alice Chen	alice@lab.ai	NLPLab	C4	0.8	ODC
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DiffNet	Carol Diaz	carol@lab.ai	VisionLab	COCO-2024	0.3	CC-BY-NC

Problem 1: This Data is Broken

- What is wrong in storing it this way?

Scenario 2: Deleting the DiffNet row.
Now Carol Diaz does not exist.

model	researcher	email	lab	dataset	sz_tb	license
GPT-X	Alice Chen	alice@lab.ai	VisionLab	ImageNet-22k	1.2	CC-BY
GPT-X	Alice Chen	alice@lab.ai	VisionLab	LAION-5B	5.0	CC-BY
LLaMA-4	Alice Chen	alice@lab.ai	NLPLab	C4	0.8	ODC
LLaMA-4	Bob Nkosi	bob@lab.ai	NLPLab	C4	0.8	ODC
DiffNet	Carol Diaz	carol@lab.ai	VisionLab	COCO-2024	0.3	CC-BY-NC

Problem 1: This Data is Broken

- What is wrong in storing it this way?

Scenario 3: Cannot add a new lab until it has at least one researcher in the same row.

model	researcher	email	lab	dataset	sz_tb	license
GPT-X	Alice Chen	alice@lab.ai	VisionLab	ImageNet-22k	1.2	CC-BY
GPT-X	Alice Chen	alice@lab.ai	VisionLab	LAION-5B	5.0	CC-BY
LLaMA-4	Alice Chen	alice@lab.ai	NLPLab	C4	0.8	ODC
LLaMA-4	Bob Nkosi	bob@lab.ai	NLPLab	C4	0.8	ODC
DiffNet	Carol Diaz	carol@lab.ai	VisionLab	COCO-2024	0.3	CC-BY-NC

How do we fix it?

How do we fix it?

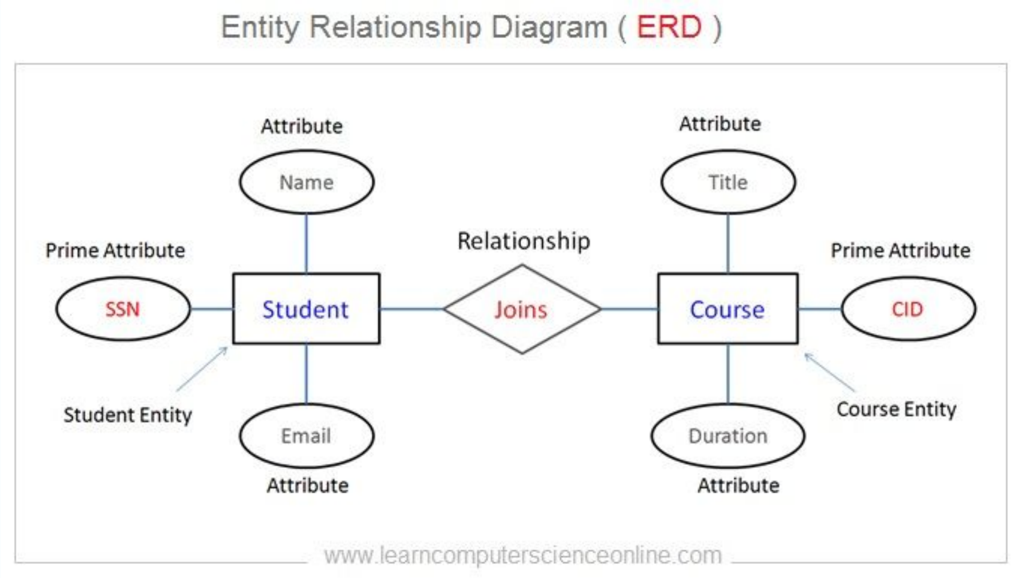
Separate the data!

Each relation should describe exactly one thing.

This is also called *normalization* – we'll get to it in Week 7!

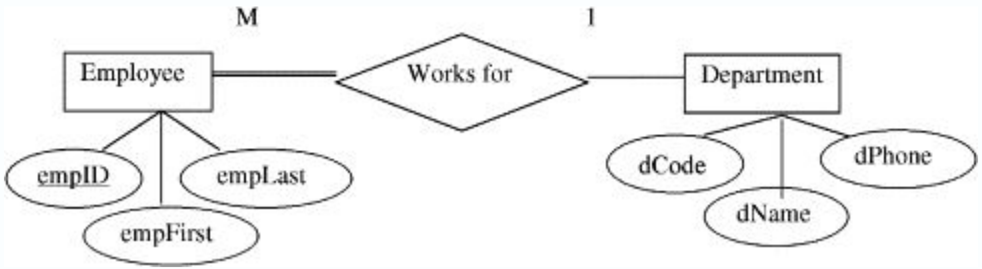
What is a DBMS?

- Collection of data
  - Entities and Relations



Entity

- An object in the real world that is distinguishable from other objects
- It has a set of attributes
  - All entities in a given entity (think of instances) share the same attributes



Entity

- For each attribute:
  - We identify a *domain* of possible values
    - E.g. a name might be the set of 20 character strings
    - A company might rate an employee from a range of 1 through 10

```
CREATE TABLE Employee (
     employee_id INTEGER PRIMARY KEY,
     name VARCHAR(20),
     rating INTEGER CHECK (rating BETWEEN 1 AND 10)
);
```

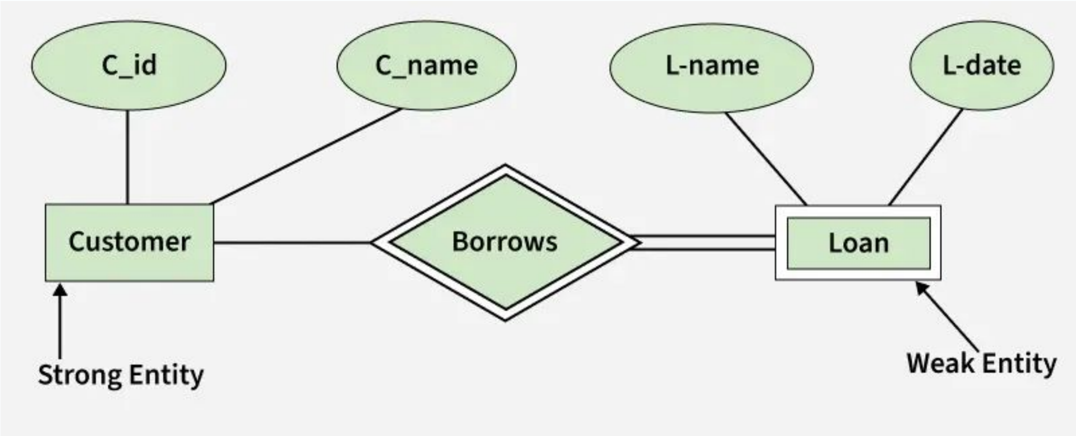
Entities and Keys

- For each entity set, we choose a *key*
  - A key is a *minimal* set of attributes whose values uniquely identify an entity in a set (think of hash values)
  - There can be more than one **candidate key**, we choose one as the **primary key**

```
CREATE TABLE Employee (
    employee_id INTEGER PRIMARY KEY,
    name VARCHAR(20),
    rating INTEGER CHECK (rating BETWEEN 1 AND 10)
);
```

Entities and Keys

- A *weak* entity is one where you cannot define a key using the entity's attributes. In this case, you need another entity to help uniquely identify an entity.



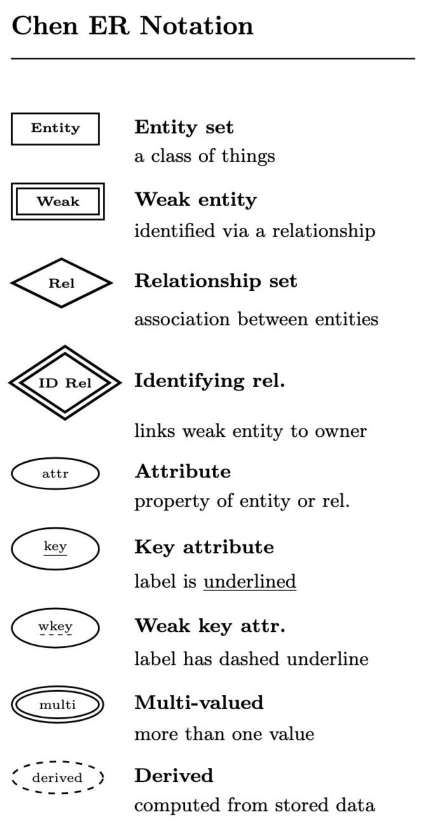
Entities and Keys

- A *weak* entity is one where you cannot define a key using the entity's attributes. In this case, you need another entity to help uniquely identify an entity.

```
CREATE TABLE Dependent (
    employee_id INTEGER NOT NULL,
    dependent_name VARCHAR(20) NOT NULL,
    relationship VARCHAR(20),

    PRIMARY KEY (employee_id, dependent_name),

    FOREIGN KEY (employee_id)
        REFERENCES Employee(employee_id)
        ON DELETE CASCADE
);
```

Entities in ERD Diagrams

- Represented by a box
  - Weak entities represented by two boxes
- Attributes represented as ovals
- Primary keys represented with an underline
- Weak keys represented with dashed underlines



Choosing Attributes

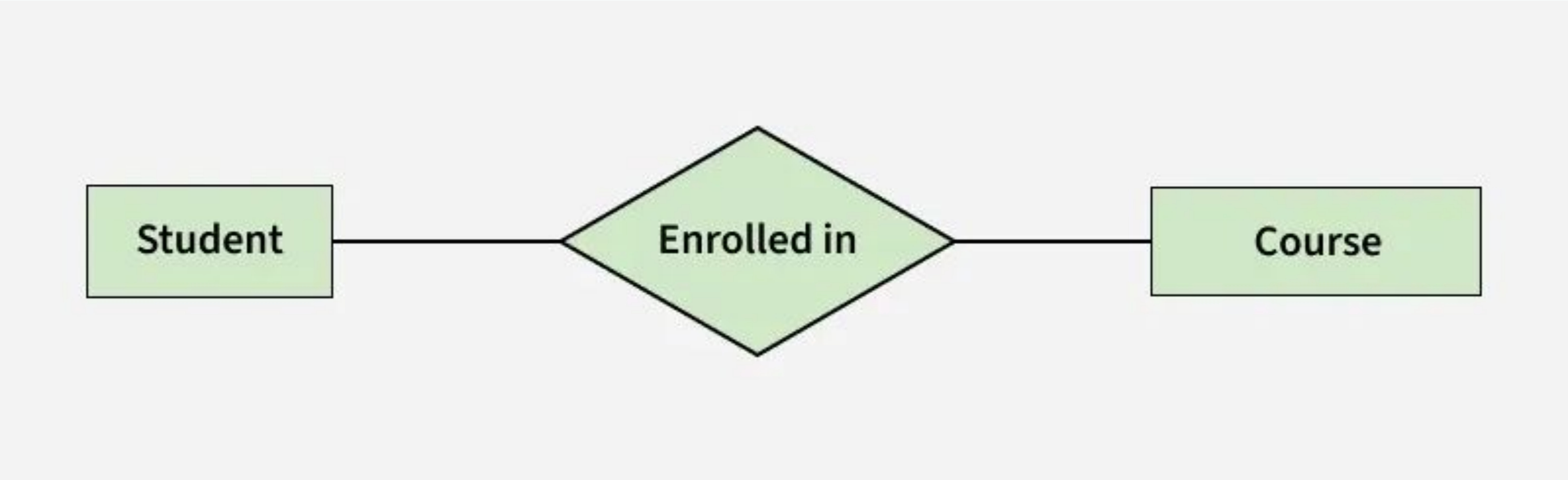
- I create an entity, Student, with the following attributes:
  - Name
  - Gender
  - Address
  - Age
  - Status (International/Domestic)
- One of these is an inefficient attribute. Which one?

Choosing Attributes

- I create an entity, Student, with the following attributes:
  - Name
  - Gender
  - Address
  - Age
  - Status (International/Domestic)
- This changes more often than we would like.
  - If an attribute can be computed using other attributes, then use a *derived* attribute instead.

Relationships

An association between two or more entities



Relationships

An association between two or more entities

```
CREATE TABLE EnrollsIn (
    student_id INTEGER NOT NULL,
    course_id INTEGER NOT NULL,
    semester VARCHAR(10),

    PRIMARY KEY (student_id, course_id),

    FOREIGN KEY (student_id)
        REFERENCES Student(student_id),

    FOREIGN KEY (course_id)
        REFERENCES Course(course_id)
);
```

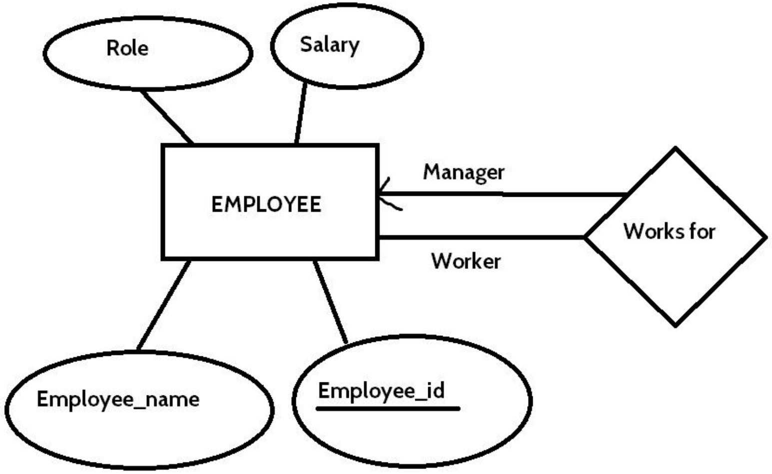
Relationships

- It can have its own attributes
- Can be thought of as n-tuples

```
CREATE TABLE EnrollsIn (
    student_id INTEGER NOT NULL,
    course_id INTEGER NOT NULL,
    semester VARCHAR(10),

    PRIMARY KEY (student_id, course_id),

    FOREIGN KEY (student_id)
        REFERENCES Student(student_id),

    FOREIGN KEY (course_id)
        REFERENCES Course(course_id)
);
```

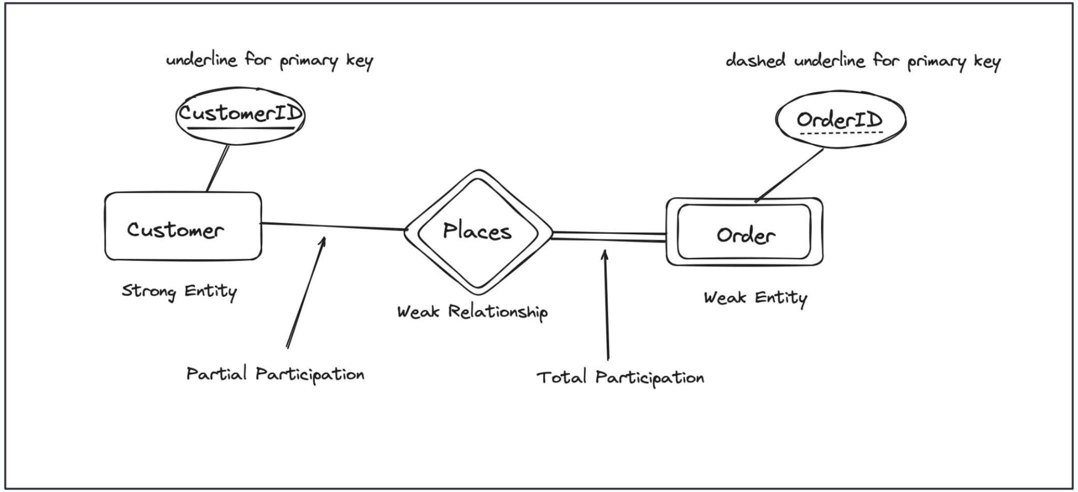
Relationships

- Entity sets that participate in a relationship don't need to be distinct!



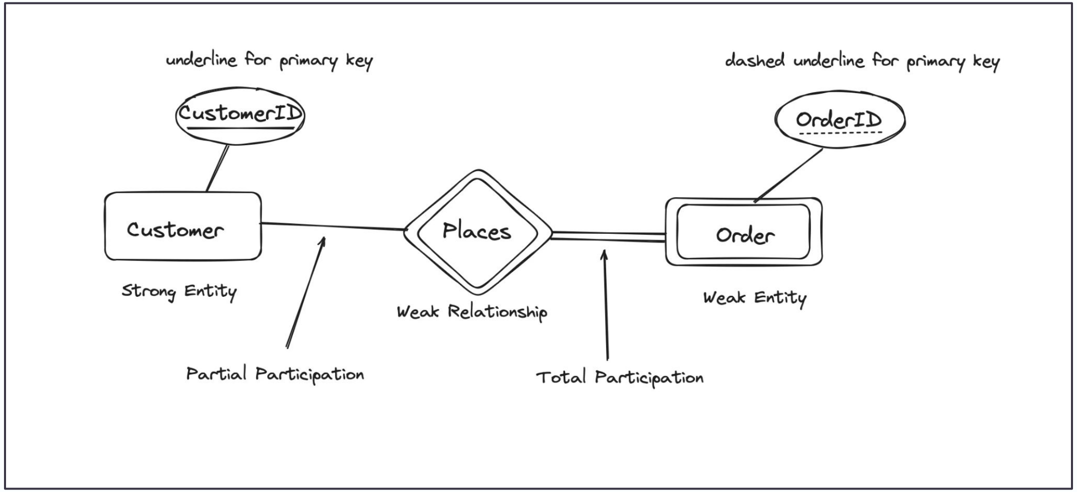
Relationships

- **Total Participation:** everyone has to participate in the relationship
- **Partial Participation:** participation is optional

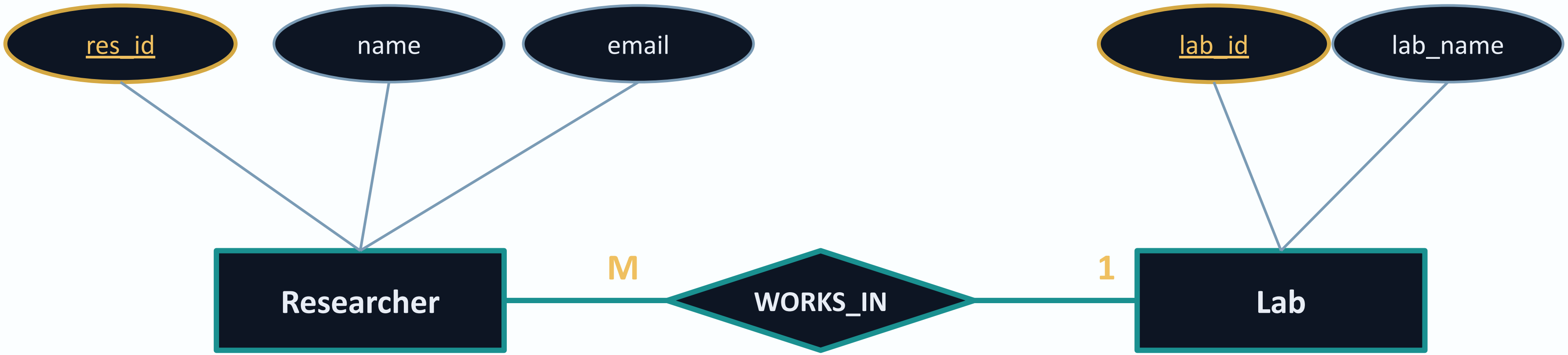


Relationships

- **Question:** can weak entities have partial participation?



Reading a Chen diagram



Handout time!

Box 1 Chen ERD Sketch ***(4 minutes)***

You have **Researcher** and **Lab** from the lecture. Add a third entity: **Project**.

- A Researcher works on **many** Projects.
- A Project has **exactly one** Lead Researcher.
- Show: entity rectangle, at least two attributes (one key, underlined), relationship diamond, cardinality.

Sketch your ERD here.

Debrief: what did you underline as the key for Project? Why not the title?

Scenario 2

You are a junior DB engineer at an AI lab. Design the schema for tracking training runs – models, datasets, researchers, compute costs, water usage per run. Your tech lead says: "Use whatever tools you want."

- Which tools would you handout to an LLM?
- Where would you not trust it?

Lecture Reflection

- Why use a DBMS?
- What does the relational model look like?

Before Next Week:

- ❑ Check tutorial 1
- ❑ Clone MarkUs Repo
- ❑ Finish PCRS prep for weeks 1 and 2

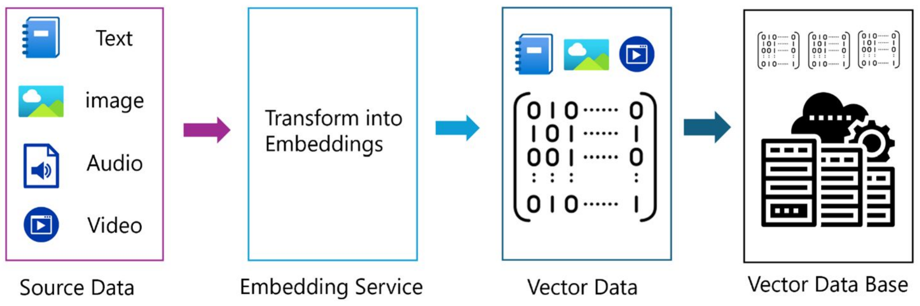
Week 2

… ERDs (IS A, Has A) and SQL continued. :)

Bonus: How Are Vector Databases Designed?

A relational database designer thinks in terms of entities, attributes, and relationships, the building blocks of ER modelling.

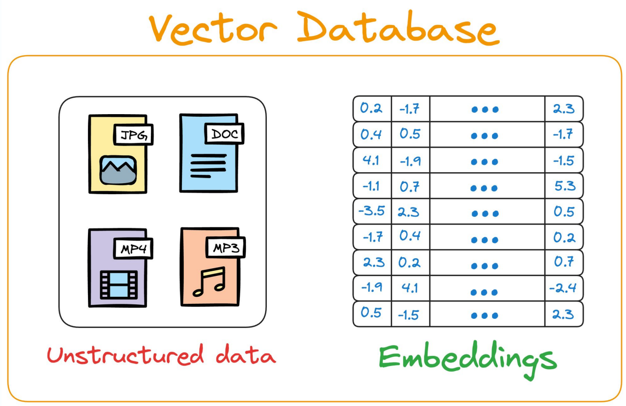
When that same designer approaches a vector database, most of those instincts still apply, but a few new questions enter the picture.



Vector Database Design

What is the **unit of meaning?**

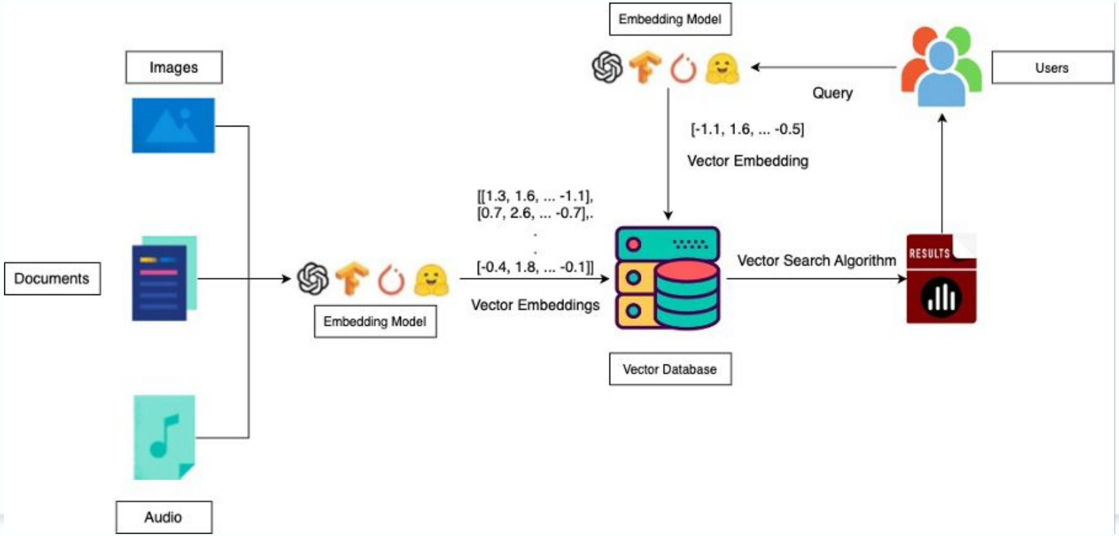
Instead of a row, the fundamental retrievable unit is often a chunk, a passage, image, or clip, small enough to be semantically coherent on its own.



Vector Database Design

How is **identity** established?

Chunks rarely have natural keys. They're often weak entities, identified only relative to a parent document or record.



Vector Database Design

What replaces the foreign key join?
Relationships between data are expressed not through explicit links but through ***proximity*** in embedding space.